

Advanced Databases

Assessment Brief

Item	Description
Module Title:	Advanced Databases
Module Code:	B9IS125
Module Lecturer (s):	Kunwar Madan Luciana Nascimento
Programme/Cohort:	MSc in Cybersecurity MSc Information Systems with Computing
Method of Assessment:	Project/technical report
Percentage (%) Weighting:	70%
MIMLOs being assessed:	1, 2, 5
Assessment Number:	2
Individual/Group:	Group
Issue Date:	01/05/2026
Submission Date:	FINAL UPLOAD AND DEMO: WEEK 12
Feedback Date:	4 weeks after submission
Feedback Strategy:	Moodle

Every student must submit their assignment on Moodle on or before the due date.

Assignments *will not be accepted via Email.*

Assignment components:

CA2 - Completed T-SQL (ALL SQL FILES MUST BE SUBMITTED)		100
	Table Creation and 5 rows of data (create and insert sql statements)	20
	4 Stored Procs (at least 2 for XML)	16
	Two Views	10
	Two Triggers	12
	Innovation	6
	Structure (Table of contents, Conclusion, Bibliography)	6
	Demonstration (in class, week 12): Connect to Database and run the SQL scripts for creating a Stored Procedure, a View with XML field retrieval, and a Trigger. Each group member should explain one SQL script and answer questions around it.	30

Assignment details:

***** PART B – IMPLEMENTATION *****

Completed Database System and demo (70 Marks) – **Submit ALL SQL Files**

***** .bak files are not acceptable here, Transact SQL must be both readable and runnable.**

Develop a **MS SQL Server** database system based on the schema developed as part of the assessment.

Your database **must include** the following:

- Create all of the normalised tables using SQL. Show your SQL scripts and the finished tables. **Tables with relationships using some referential integrity constraints.** There should be a minimum of five substantial tables and any number of smaller tables that you require to implement your database. Tables should be linked using an appropriate method in **MS SQL Server**. Each table must have a minimum of **5 test records per table**. Enter sample data in all tables. In your report explain which referential integrity constraints are used and why. **(20 marks)**
- Write at least **4 (four) Stored Procedures with Parameters** (at least 2 for XML) that show your understanding of SQL. SQL statements that should be used should include: **Select/Update/ Delete, From, Where, And, Count, Ascending.** The queries that meet business requirements and must include the following features:

 - i. Use of **JOIN between two or more tables** as required.
 - ii. Use of **GROUP BY with HAVING.**
 - iii. Use of **custom (User Defined) SQL functions.**
 - iv. **Developing XML** with appropriate elements using relational fields.
 - v. **Modifying/Searching** data in a field of **XML data type**.

(16 marks)
- **Two triggers** to demonstrate the implementation of business rules. **(12 Marks)**
- **Two Views** to demonstrate the development of virtual tables. How Views would be helpful to generate customised View of your data? What type of clients would need that data? TWO (2) of the queries should **JOIN, Order By**. TWO (2) or more tables together. At least one query must make use of **GROUP BY**. At least one query must **retrieve data** logically **from a field with XML data type** as well as data from fields from other data types. **(10 Marks)**
- Report must be well-structured with Conclusions and Bibliography. **(6 marks)**

Document Innovation (6 marks)

Outline the use of any other features to enhance the usability of your database system. Write a paragraph on innovation and how it is useful. ***This should be a MS Word Document included with the SQL Files.***

Demonstrate in Person:

Connect to Database and run the SQL scripts for creating a Stored Procedure, a View with XML field retrieval, and a Trigger. Each group member should explain one SQL script and answer questions around it.

(30 marks)

Group Management

Refer to the College's Group Work Policy, [QAH B.5.2.2.2](#).

Group Sizes:	3
Group allocation:	Students choose
Grading approach:	Single Grade for the whole group . However, individual grades depend on a satisfactory performance during the Q&A in the demo.
Alternative assessment:	Learners with PMCs that require an alternative assessment must log a ticket for consideration.

Alternative Assessment

- Any individuals who cannot participate in a group must undertake the task as an individual report.

Assessment criteria

Find below the Evaluative Criteria/Performance Level Descriptors per assessment component.

Grades of 80+ are subject to meeting the grade band criteria and achieving the milestones published on Moodle, if any.

Implementation: Database & T-SQL (70 Marks)					
Criteria / Mark	<40	40-49	50 - 59	60 – 69	70+
Tables & Relationships	< 5 tables OR no data or incorrect datatypes or too many errors to run. OR FK and PK referential integrity is NOT maintained at all, i.e. FKs applied after insert or NO FKs.	5 Tables. Data inserts for <=2. Others have errors or not enough rows to insert. OR FK and PK referential integrity is NOT maintained >=2 Tables.	5 Tables. Data inserts for <=3. Others have errors or not enough rows to insert. OR FK and PK referential integrity is NOT maintained in >=1 table.	5 Tables. Data inserts for <=4. 1 insert has errors or is missing or has less than 5 rows for the table. AND FK and PK referential integrity is maintained	5+ Tables. All inserts run error free. There are 5+ rows of data per table and the FK and PK referential integrity is maintained.
Two Triggers	None supplied or code does not run	Only 1 Trigger supplied. Not clear what it's purpose is in terms of business rule.	Two Triggers supplied. Both of the same type, e.g. Instead Of Triggers. Not clear what the purpose is in terms of business rule for at least one of them.	Two Triggers supplied. Two different types. Not clear what the purpose is in terms of business rule for at least one of them.	Two Triggers supplied either the same type or different. Both clearly meeting a Business Rule Each. Both run and achieve this.

Two Views	Not supplied or do not create	Create but do not return any data. Or supplied and return data, make no sense to the system.	Only 1 supplied. It runs and returns data. Makes good business sense to the system	Both supplied, both run and only 1 returns data. Make good business sense.	Both supplied. Make good business sense. They create and return data.
Parameterized Stored Procedures (At least 4)	Not supplied or do not compile/execute	3 < supplied with parameters. Compile and run.	4< supplied with parameters. Meet the brief in terms of Clauses to be used. Compile and run. Not all return data however at least 2 do.	All 6 supplied with parameters. Meet the brief in terms of Clauses to be used Compile and run. Not all return data however at least 3 do.	All 6 supplied with parameters. Meet the brief in terms of Clauses to be used Compile and run. All return data as expected.
Report Layout	Very poor	Some	Missing two	Missing	Well laid out,

Demo	Layout, missing nearly all elements.	headings around each section. Some elements not done or done to a low standard.	pieces such as Conclusion or Requirements. OR sections merged or confused in together whilst others are very distinct.	Conclusion or References. Requirement , Scope, and other elements all done to a high standard.	table of contents. Flows well. Good grammar and spelling. All sections included with screenshots Examples.
	Did not attend/ SQL did not run / could not answer questions on the document and project.	Some SQL runs, some minor errors and at least 2 questions answered well.	Most SQL runs, some minor errors and at least 3 questions answered well.	All SQL runs, no errors. Majority of questions answered well, stumbled or not sure of 1.	All SQL runs, no errors. ALL questions answered very well, showing a clear knowledge of the subject matter.

Innovation	Not supplied.	Very simple attempt at this < = 4 lines. Primarily referencing things already outlined in the brief. No explanation of what innovation is.	Very simple attempt at this < = 10 lines. Primarily referencing things already outlined in the brief. Some small additions outside of this. No explanation of what innovation is.	A paragraph on this. Mixture of items in the brief and others. Includes an explanation of what innovation is.	A well written paragraph outlining the use of any other features to enhance the usability of your database system, what innovation is and how it is useful.
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General Assessment Submission Requirements for Students:

1. Online assignments must be submitted no later than the stated deadline.
2. All relevant provisions of the Assessment Regulations must be complied with.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please log a service desk ticket <https://servicedesk.dbs.ie/HEAT> . Please supply a medical cert or other required documentation.
4. Students are required to retain a copy of each assignment submitted, and the submission receipt.
5. There is a penalty of 2 marks per day for late submissions.
6. Assignments that exceed the word count will be penalised.
7. Students are required to refer to the assessment regulations in their Student Guides and on the Student Website.
8. Dublin Business School penalises students who engage in academic impropriety (i.e., plagiarism, collusion and/or copying). Please refer to the attached referencing guidelines for information on correct referencing.

What is referencing and why is it necessary?

Please follow this link to the Harvard Style Referencing Guide - all referencing is required in this format.

<https://libguides.dbs.ie/skills/referencing/harvard>